CHAPTER 1.
ROTATIONAL DYNAMICS

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- Recap:

MECHANICS → A branch of physics dealing with particles (or objects) either at rest or in motion.

STATICS → Study of objects at rest

KINETICS → Study of moving objects

Kinematics

- In this we study of motion without considering the causes
- Eg. distance., displacement, speed, velocity, acceleration, etc.

Dynamics

- Study of motion along with its causes
- To study distance, velocity etc. along with the force or torque which has caused the motion of that object.
- Eg. Momentum, force, energy, power, etc.
- Branches of Dynamics: Fluid Dynamics, Aerodynamics, etc.

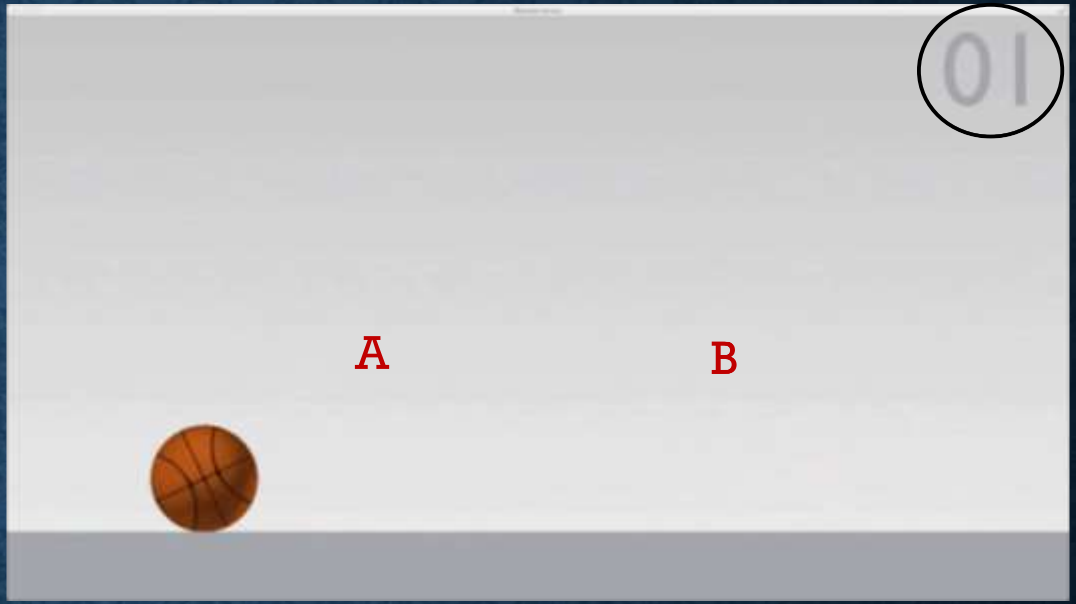
To understand the terms : Motion, Kinematics and Dynamics let us see next slide

- “Motion”? “It is a change in the position of an object with time.”
- Example, a rolling ball
- In this as the ball moves from Point A

to Point B , we can also observe the

changes in time given at top right

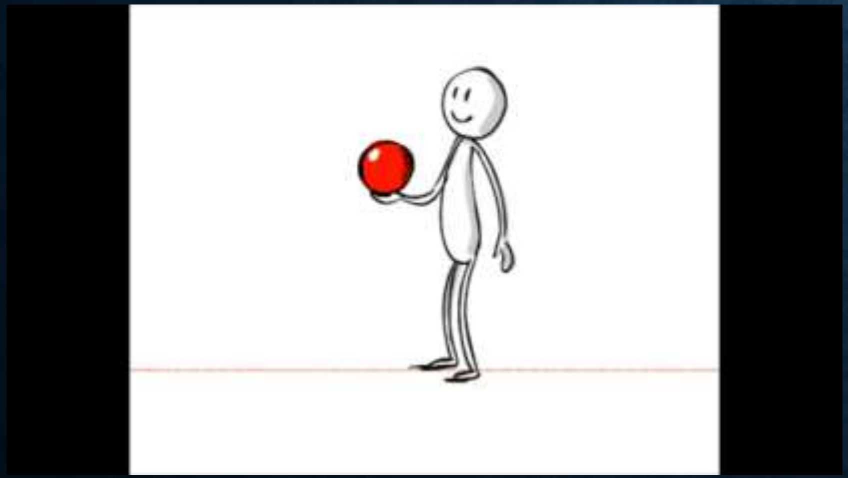
corner



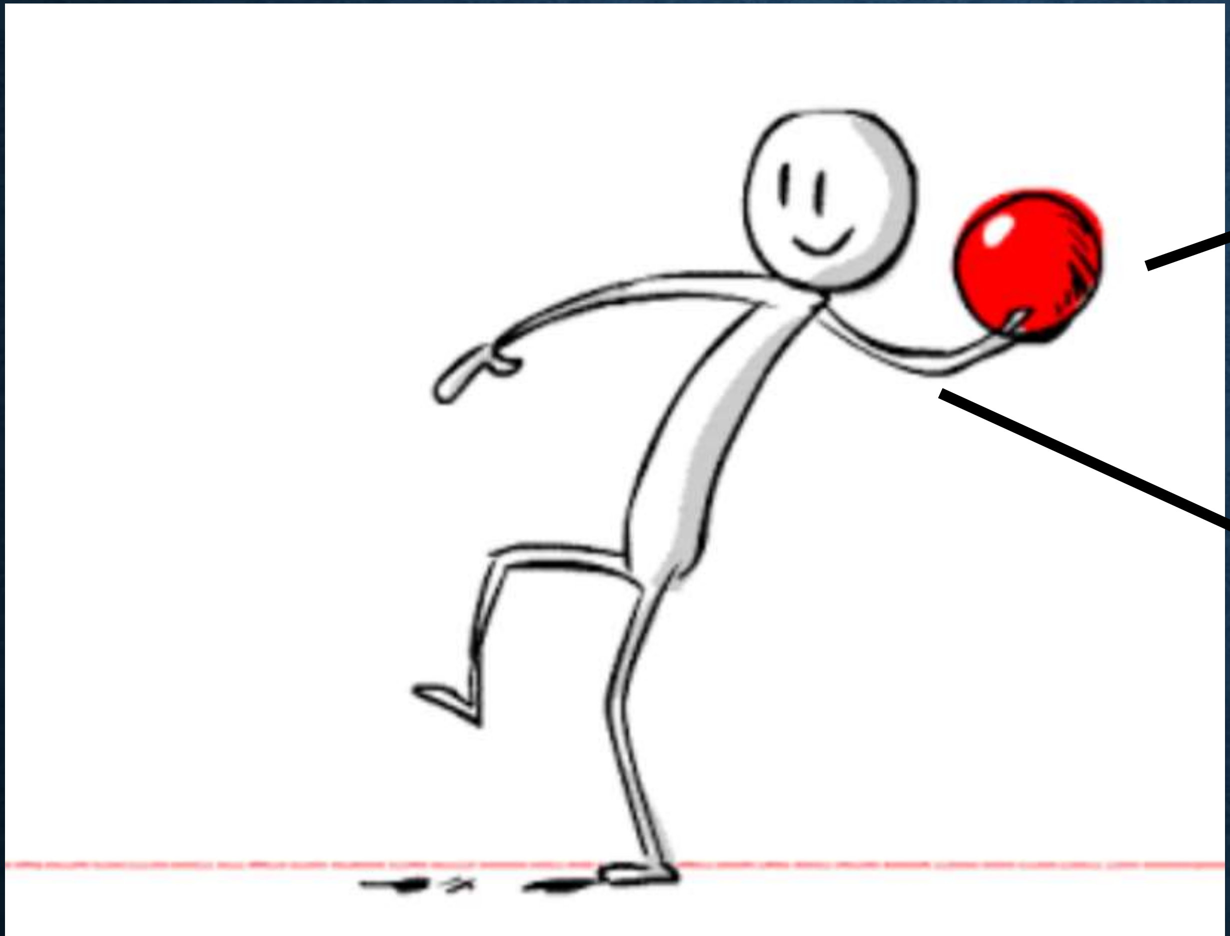
time

Changing positions

Kinematics: study motion without cause
Dynamics: study motion with cause



Consider what is happening here,



Throwing a ball (Kinematics-distance velocity)

Force applied by his hand is the cause for above motion (Dynamics)

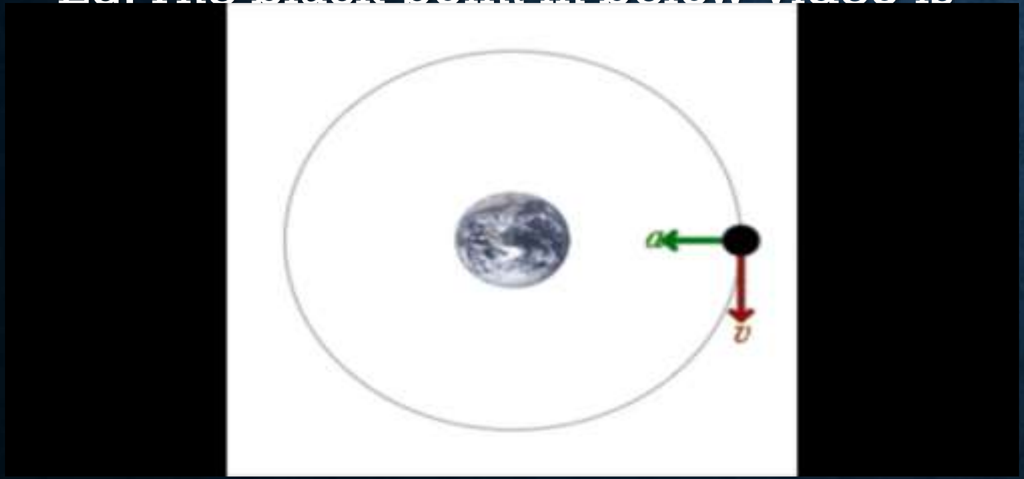
So when we say Dynamics, we will study this motion (throwing ball) along with its cause (applied force)

In class 11, we studied motion but along a line i.e. a linear motion, here we will study rotational motion but with its causes. Hence the name Rotational Dynamics.

Before we go ahead, understand the difference between Rotational and Circular Motion.

Circular: object moving in circle around a point or another object.

Eg. The black point in below video is



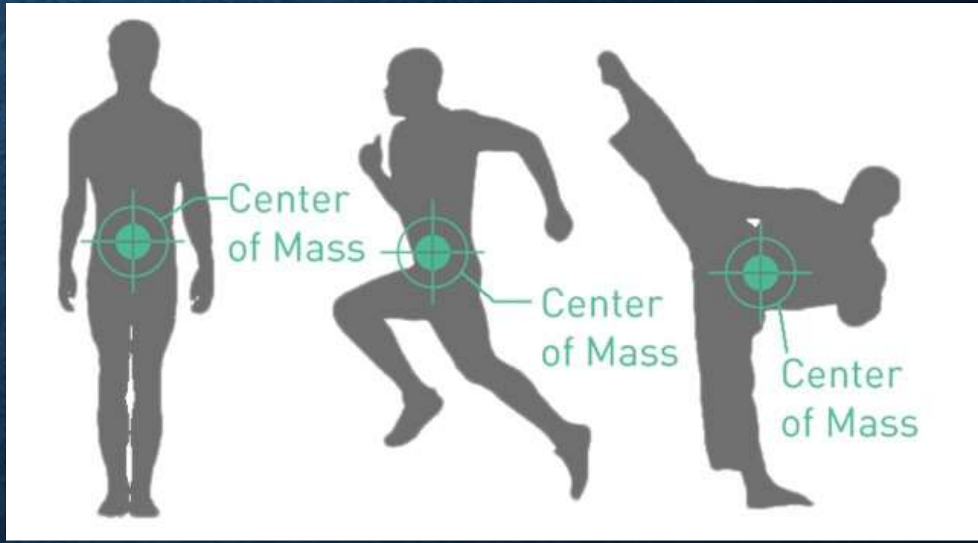
Rotational: object is "turning" around itself eg. Rotation of earth around itself

- Do you recall? (from textbook)

1) Circular motion? Motion of an object along a circular path.

2) Centre of Mass: It is an imaginary point that we consider inside a body where whole mass of the body is supposed to be concentrated.

- It gets shifted as per our position.
- You can balance objects at their point of Centre of Mass , eg. a baseball bat.



3) Kinematic Equations:

$$v = v_0 + at$$
$$v^2 = v_0^2 + 2a\Delta x$$
$$\Delta x = v_0 t + \tfrac{1}{2}at^2$$

4) Real Forces:

- Arises due to actual interaction between objects, or simply put whose source and effect both can be seen.
- Eg. Throwing a ball

Pseudo Forces:

- only effect can be seen or felt, source many times unknown.
- Eg. Inside an elevator, you feel lighter when it is moving down and heavier when it is going up.

5) Rigid Body:

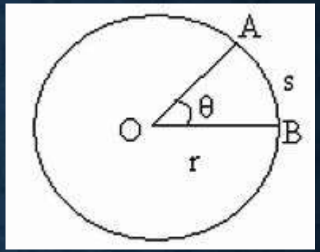
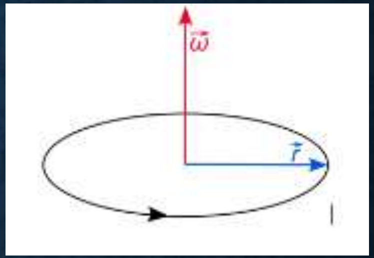
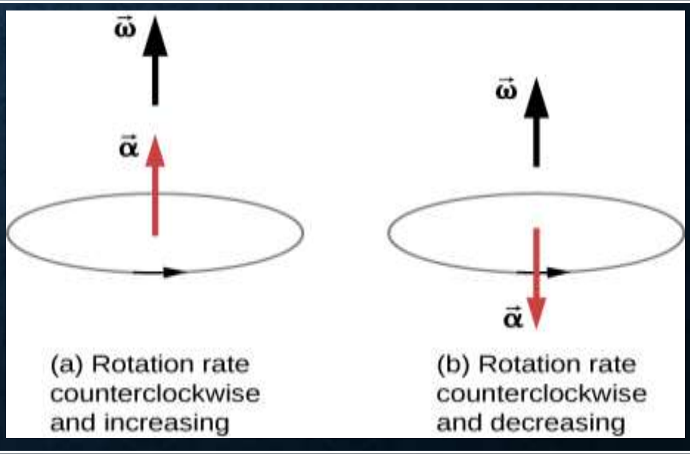
- Idealisation of solid body
- Distance between any two points inside a rigid body remains constant with time.

CONTENTS:

The topics in present chapter are as follows:

1. Horizontal Circular Motion: a) Kinematics and Dynamics. b) Its application.
2. Vertical Circular Motion
3. Moment of Inertia, M.I. (analogous to mass) and Torque in terms of M.I.
4. Radius of Gyration
5. Theorem of parallel and perpendicular axis
6. Angular Momentum
7. Rolling Motion

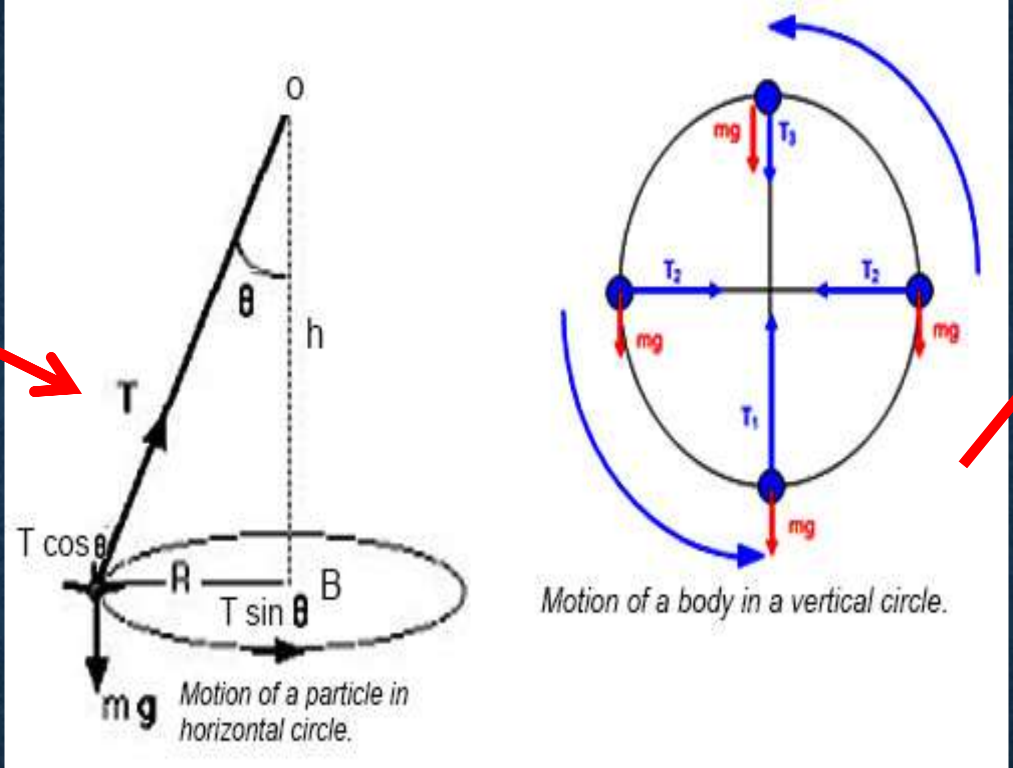
Analogies between Linear and Rotational Motion:

Linear (or Translational)	**Rotational**
Displacement of object on straight line- (x)	Displacement in circular motion- <u>Angular Displacement</u> (θ theta) Unit: Radian (rad) 
Velocity along straight line- (v) $V = \frac{dx}{dt}$	Velocity along a circular motion- <u>Angular Velocity</u> (ω omega) Unit: (rad/m) $\omega = \frac{d\theta}{dt}$ 
Acceleration along a straight line- (a) $a = \frac{dv}{dt}$	Acceleration in circular motion- <u>Angular Acceleration</u> (α alpha) Unit: (rad/m^2) $\alpha = \frac{d\omega}{dt}$ 

In textbook, The first part of lesson talks about topics considering

Horizontal Circular motion: eg. A pendulum

- UCM
- Non-UCM
- Centripetal & Centrifugal Force and Its applications



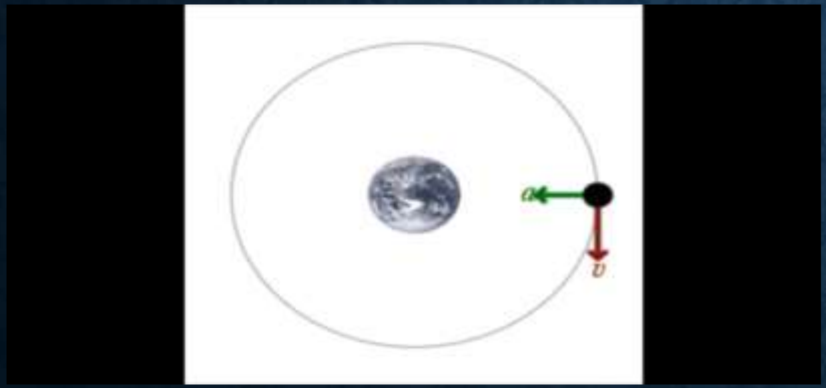
In second half of lesson we discuss

vertical circular motion.

Eg. rolling wheel

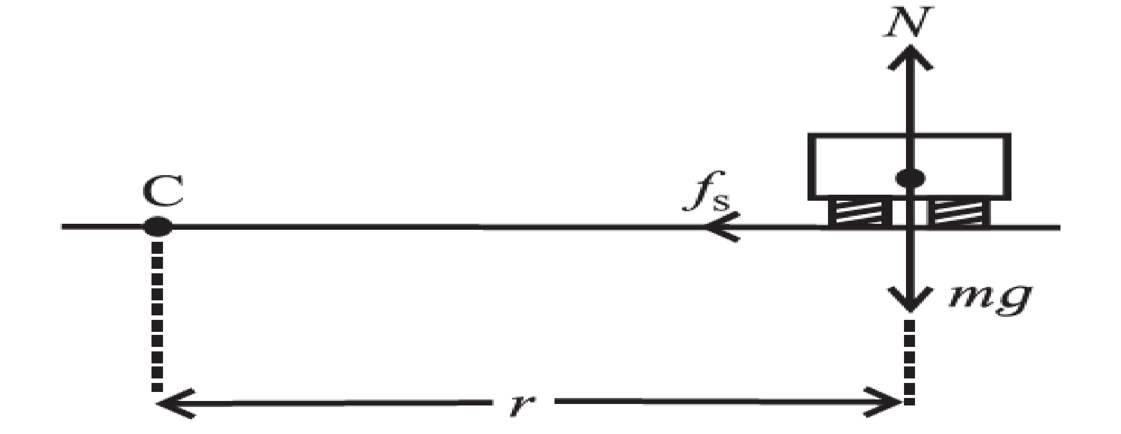
It is possible to discuss UCM, Non-UCM , Centripetal and Centrifugal force for Vertical Circle also, it is simply not being discussed here.

Uniform Circular Motion	Non Uniform Circular Motion
Eg. Fan moving with constant speed	Eg. When Fan is switched ON its speed goes on increasing (when put OFF speed goes on decreasing)
Speed of particle remains constant	Speed of particle either increases or decreases
Acceleration responsible is "Centripetal Acceleration" also called as "radial acceleration" whose value remains constant. $(\bar{a}_r)$	Value of radial acceleration goes on changing$(\bar{a}_r)$
Magnitude of $\bar{a}_r$ remains constant.	Magnitude of $\bar{a}_r$ is not constant.
Angular acceleration (α) is zero	For increasing speed, α is along the direction of ω. For decreasing speed, α is opposite to the direction of ω
Involves only Centripetal force	With Centripetal force involves additional forces due to non zero α

Centripetal Force	Centrifugal Force
The acceleration required for circular motion is provided by Centripetal force	Centrifugal force is the pseudo force that arises due to acceleration of frame of reference.
Frame of reference is steady	Frame of reference is accelerated.
Always directed towards the centre	Always directed away from centre
It is real force.	It is non-real force. But we cannot call it "imaginary" because its effects are seen
Eg. Satellite orbiting around the earth 	Eg. Mud flowing away from tyre 
Resultant force, $\mathrm{F} = - \mathrm{m}\omega^2\bar{r}$	Resultant force, $\mathrm{F} = + \mathrm{m}\omega^2\bar{r}$

APPLICATIONS OF UCM

1) Vehicle along a horizontal circular track.



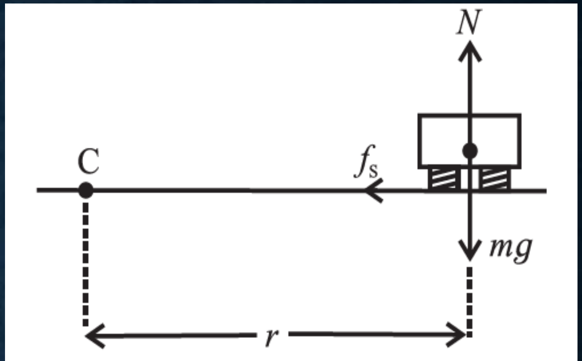
(Vertical Section)

Forces acting on car:

1) Weight (mg) downward
2) Normal reaction (N) upward
3) Static friction (f_s) sideways inward ⟶ provides Centripetal force

When we consider moving car as our frame of reference, centripetal force is balanced by centrifugal force

- Aim: to find equation of v



To find v, we shall consider following two conditions,

Normal reaction(N) balances weight (mg) of car i.e. N = mg ------ (1)

Static force (f_s) is balanced by centrifugal force, i.e. $f_s = + m\omega^2 r$ ------ (2)

If we divide (2) by (1) we get $\therefore \frac{f_s}{N} = \frac{r\omega^2}{g} = \frac{v^2}{rg}$ since v = r ω.

From the above box,
we can see a direct proportionality in f_s and v, also if we consider the equation for static friction, the first term equals μ.

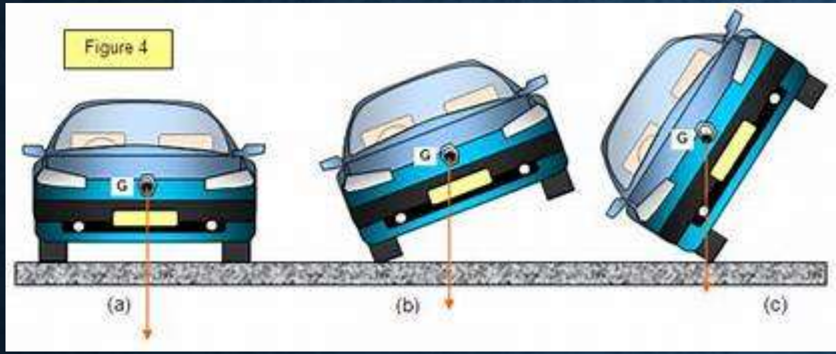
For a given track, radius r is constant. For given vehicle, $mg = N$ is constant. Thus, as the speed v increases, the force of static friction f_s also increases. However, f_s has an upper limit $(f_s)_{max} = \mu_s . N$, where μ_s is the coefficient of static friction between road and tyres of the vehicle. This imposes an upper limit to the speed v.

At the maximum possible speed v_s, we can write

$$\frac{(f_s)_{max}}{N} = \mu_s = \frac{v_{max}^2}{rg} \quad \therefore v_{max} = \sqrt{\mu_s rg}$$

By far we discussed this application by considering the vehicle as a point, but in reality when we have to consider the 4 tyres of car and their respective weights, normal reactions and also frictional forces and centripetal forces.

In such case as the forces are not balanced, torque is produced causing the vehicle to "topple"



- In circular motion car toppled because frictional force is not balanced by centripetal force.
- This can happen for two wheeler also.

When two wheeler is moving on a circular track, the rider must keep his bike at certain inclination to ground to avoid toppling.

Eg. While taking turn the bike racer tilts himself.

- Talking about a driving bike or car on a circular path, takes us to our next application

2) Well of Death (मौत का कुआ) : It is a cylindrical wall on which stuntmen drive car or bike

Again, we consider the forces acting on the car:

1) Weight (mg)----- vertically downward
2) Normal reaction (N) ---- acting horizontally towards centre
3) Static friction (f_s) vertically upwards between the tyre and wall (static because it must prevent downward slipping)

Now in this case the two cases are,
Weight (mg) is balanced by static force (f_s) i.e. $f_s = mg$ ------ (1)
Normal reaction (N) provides Centripetal force i.e. $N = + m\omega^2 r$ ------ (2)

(Notice that in previous application frictional force was providing centripetal force while here normal reaction is doing it)

1. Vehicle along circular track

For a given track, radius r is constant. For given vehicle, $\boxed{mg = N}$ is constant. Thus, as the speed v increases, the force of static friction f_s also increases. However, f_s has an upper limit $(f_s)_{max} = \mu_s . N$, where μ_s is the coefficient of static friction between road and tyres of the vehicle. This imposes an upper limit to the speed v.

At the maximum possible speed v_s, we can write

That means when f_s balances Centripetal force, we get v_{max}

$$\frac{(f_s)_{max}}{N} = \mu_s = \frac{v_{max}^2}{rg} \therefore v_{max} = \sqrt{\mu_s rg}$$

2) Well of Death:

Normal reaction N is thus the resultant centripetal force (or the only force that can balance the centrifugal force). Thus, in magnitude,

$$N = mr\omega^2 = \frac{mv^2}{r} \text{ and } \boxed{mg = f_s}$$

Force of static friction f_s is always less than or equal to $\mu_s N$.

$$\therefore f_s \le \mu_s N \therefore mg \le \mu_s \left(\frac{mv^2}{r} \right)$$

$$\therefore g \le \frac{\mu_s v^2}{r} \therefore v^2 \ge \frac{rg}{\mu_s}$$

when N balances Centripetal force, we get v_{min}

$$\therefore v_{min} = \sqrt{\frac{rg}{\mu_s}}$$

(Notice that in first application frictional force was providing centripetal force while in second normal reaction is doing it)

- Again throughout the derivation we assumed the car or bike to be a point particle, but this is not the case in reality.
- Also, during revolutions the two wheeler is in fact aligned at a particular angle to the wall, this is to avoid toppling. Thus the bike is never perfectly 90 degree with horizontal.
- This gives us following interpretations,

1) If the angle between bike and wall is $\theta = 90°$ then it imposes lower limit on speed.
2) If the angle is $\theta = 0°$ (unbanked road, first application) this imposes upper limit.
3) For $0° \leq \theta \leq 90°$, the turning speed has both upper and lower limit.

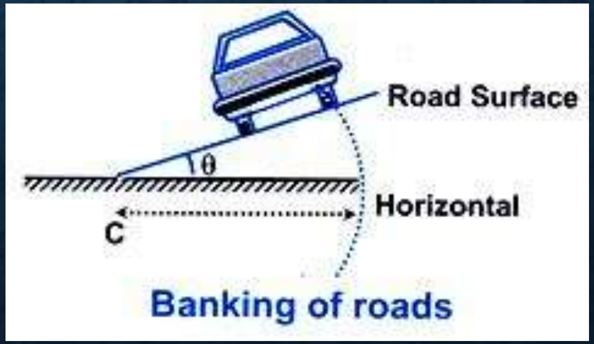
The θ here is in fact Banking Angle, which takes us to our next application.

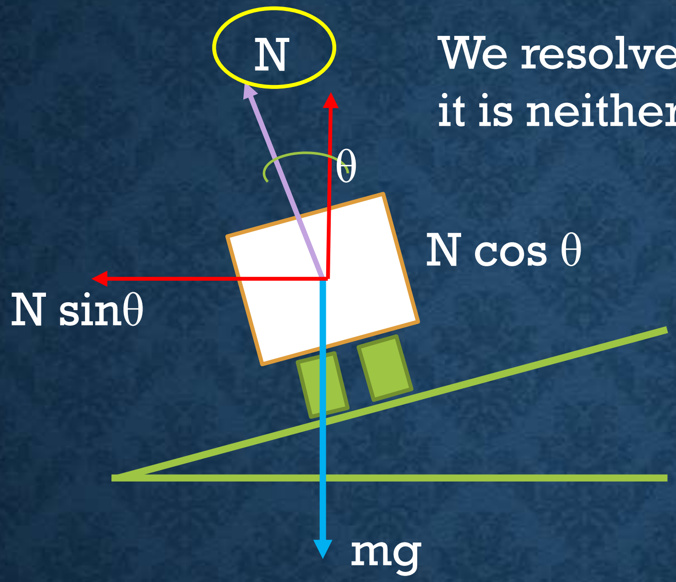
3) Vehicle on a banked road

- As we discussed in previous application, when taking a turn, the static force (f_s) between tyre and road surface provides the necessary centripetal force, i.e. f_s balances the centrifugal force(avoiding the vehicle to skid away from track)
- But this f_s has upper limit of $\theta = 0°$, also its value changes for non-uniform surfaces (say, when you have a rough road). Therefore, we cannot depend only on this frictional force, between tyres and road surface, to keep the vehicle on track when taking turn…! Hence, it is necessary to "bank" the road surface.
- Banking of road means the outer edge of curved roads are slightly lifted or we can say, the surface of roads are made such that they are tilted at some angle to horizontal.
- The speed limit on road is determined by this

Banking angle. The racing tracks are banked at

Higher angles than road through ghats.





We resolve this N into its components because it is neither along Y axis or along X axis

Considering the car to be a point and ignoring non-conservative forces like friction, air resistance etc. we make the list of forces involved.

1) Weight (mg) vertically downwards
2) Normal reaction (N) perpendicular to the surface of road, resolved into its components

From figure, we get, $N \cos\theta = mg$ ------- (1) and $N \sin\theta = mv^2/r$ ------- (2)

Dividing (2) by (1) we get, $\tan\theta = v^2/rg$ i.e.

$\theta = \tan^{-1}(v^2/rg)$ Also, $v = \sqrt{r\,g\tan\theta}$

- CASES:

1) Most Safe Speed:

The equation $v = \sqrt{r\,g\tan\theta}$ gives us the value of most safe speed on any road (not minimum or maximum speed)

2) Banking Angle:

This helps us in designing a road and is given by $\theta = \tan^{-1}(v^2/rg)$

3) Speed limits:

When started this discussion we ignored the forces due to friction, but in reality we have to consider them, therefore our formula for v slightly changes.

Let us consider same diagram but now with friction forces.

Forces involved here are,

i. Weight(mg) vertically downward

ii. Normal reaction perpendicular to road surface

iii. Frictional force along the road surface.

N and f_s are further resolved into its components.

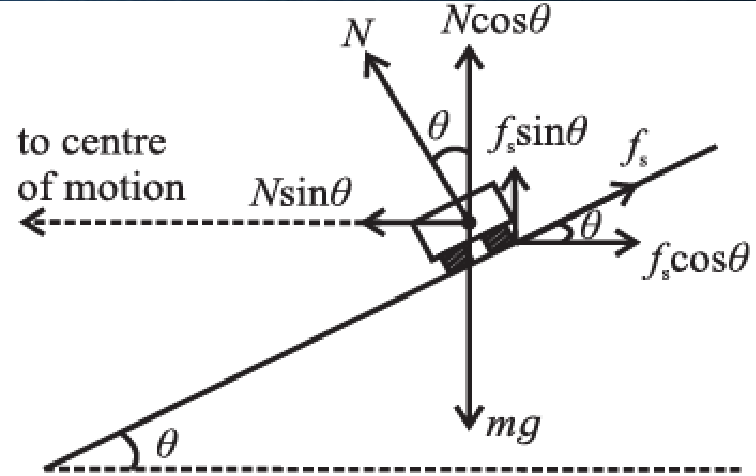


Fig 1.7: Banked road : lower speed limit.

(Derivation in word document)

Case c: When speed v_1 is less than safe speed i.e. $v_1 < \sqrt{rg\tan\theta}$ and $\frac{mv_1^2}{r} < N\sin\theta$

$$(v_1)_{min} = v_{min} = \sqrt{rg\left(\frac{\tan\theta - \mu_s}{1 + \mu_s\tan\theta}\right)}$$

Therefore, in this equation, for $\mu_s \geq \tan\theta$ then $v_{min} = 0$

True for ROUGH ROADS, which are banked at smaller angles

(we did (c) case to find out how lower one can go than safe speed, so as to still be able to take turn)

Case d: When speed v_2 is more than safe speed i.e. $v_2 > \sqrt{rg\tan\theta}$ and $\frac{mv_2^2}{r} > N\sin\theta$

$$(v_2)_{max} = v_{max} = \sqrt{rg\left(\frac{\tan\theta + \mu_s}{1 - \mu_s\tan\theta}\right)}$$

Therefore, in this equation, for $\mu_s = \cot\theta$ then $v_{max} = \infty$
But maximum value for $\mu_s = 1$, therefore for $\theta \geq 45°$, $v_{max} = \infty$.

What they mean here is that if your road is heavily banked i.e. you have banking angle to be 45 degree you can literally drive with infinite speed. NOT POSSIBLE IN REALITY !!

Case e:

- Therefore we have our next case where we take friction to be zero i.e. $\mu_s = 0$, this takes us back to the equation of safest speed, where we do not consider (as in take help) of friction, $\mathbf{v} = \sqrt{r\, g \tan\theta}$